

DR1: Can a Cheap Nominator Find the Load-Bearing Deletion? A Smallest Working Example

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Deletion-Repair Nomination – probe DR1 **Status:** smallest working example. NO_GO on both preregistered hypotheses. **Date:** 2026-07-24

Abstract

A discovery-shaped move can be modelled as: hold a representation R that works on a child task α , delete an over-specification $D \subseteq R$, repair with r , and thereby cover a parent task ω that R could not. Relativity and attention both fit: absolute simultaneity was free at low velocity and fatal at high; recurrence was free at short sequence length and fatal under a parallel budget. In **both** canonical cases the repair already existed – Lorentz had published the transformations in 1904, attention was standard by 2015 – so what was hard was the *deletion*, not the replacement.

DR1 asks the smallest question that follows: can an execution-free **nominator** rank the load-bearing deletion highly, scored against an exhaustive oracle, on toy systems where the answer is known? Two toys, one harness. TK is relativity-shaped; TT is attention-shaped.

Two hypotheses were preregistered and **both returned NO_GO**, informatively. **H1** (no single nominator wins on both toys) failed because its binary framing does not survive contact: "weakness is silent" is almost never true, since extension size grows for nearly any non-vacuous deletion. The *graded* version of the claim does hold – weakness scores 1.00 vs cost's 0.00 on TK, and cost scores 1.00 vs weakness's 0.67 on TT. **H2** (a max-disjunction is at least as good as the better single nominator) failed outright: the disjunction is **worse** than cost on TT (0.67 vs 1.00), because max after normalisation lets a high-weakness-but-useless deletion tie a high-cost-and-correct one.

A third result was not preregistered because it happened during construction: **the leakage gate built for the previous program fired on this one**. Alphabetical tie-breaking handed a *completely silent* nominator – cost on TK, with every score tied at zero – a load-bearing deletion in its top 3. That is a permitted field (the alphabet) carrying information it had not earned. It was fixed before freezing.

1. What DR1 is not

DR1 does not discover anything. Relativity and attention are already discovered; they cannot be found again, only *retrodicted*. They are the validation targets – the known top-k – not the goal. DR1's toys are miniatures of the **shape** of those moves, not of their content, and a GO here would have licensed exactly one thing: preregistering a date-cut retrodiction on a real corpus.

The vocabulary Σ is fixed throughout. Deletion-and-repair *inside* a fixed language is what DR1 mechanises. Language *extension* is not, and that is the known ceiling: special relativity is expressible in 1904's vocabulary, but general relativity required Riemannian geometry Einstein had to go and learn. Same physicist, same decade, opposite sides of the ceiling.

2. Design

Both toys expose the same five slots – a finite hypothesis space H , propositions acting as predicates over H , a child task α , a parent task ω , and a cost model – so one harness scores both.

2.1 TK – Toy Kinematics (relativity-shaped)

H is a grid of coordinate transformations at relative velocity v (units $c = 1$), indexed by a single physical dial k : $k = 0$ is the Galilean member, $k = 1$ the Lorentz member. Dilation is **not** an independent dial – it is a facet of time mixing, exactly as length contraction follows from relativity of simultaneity rather than being assumed separately.

term is negligible. Galilean and Lorentz agree to tolerance. **α cannot tell them apart** – this is the over-specification-fitted-to-the-child-task condition, and it is what makes the puzzle a puzzle.

$x' = t'$ at $v \in \{0.3, 0.6, 0.9\}$. Only the Lorentz member survives.

- α probes $v \in \{0.01, 0.02, 0.05\}$ at a small displacement, so the $v \cdot x$
- ω adds light-propagation invariance: a ray $x = t$ must satisfy

Crucially, `absolute_simultaneity` and `no_length_contraction` are entangled facets of one commitment: **neither alone frees anything**. Each scores weakness gain of exactly 0.0. The load-bearing deletion is the *pair*. A nominator that scored only singletons would miss it entirely.

`preferred_rest_frame` is the deliberate trap – an ether-shaped commitment that is droppable and *does* enlarge the extension, but does not reach ω on its own. It is the Lorentz-without-Einstein move: give up the ether, keep absolute time, get nowhere.

2.2 TT – Toy Transduction (attention-shaped)

H is a set of computation schemes over sequences, each with an access pattern and a parallel depth. α is short sequences under a loose budget, where sequential and parallel schemes both pass. ω is a parallel-depth budget that sequential schemes cannot meet.

2.3 Nominators, and the oracle

Five nominators, all execution-free – none may call `fits_omega`: weakness (extension growth), cost (best-achievable-cost improvement), disjunctive (max after per-toy normalisation), and two controls, random and size_only.

The oracle is exhaustive over $|D| \leq 2$, labelling each D valid-on- α and covers- ω . Negatives are included by construction – propositions whose deletion frees nothing – so the recall denominator means something.

3. The gate that fired during construction

Erratum E1 of the previous program recorded a defect in which a *permitted* field already contained the answer, and every audit missed it because each had asked whether the policy could reach something *forbidden*. DR1 assumed that failure mode was present until measured.

It was. cost is completely flat on TK – every hypothesis costs the same, so every candidate's cost attribution is exactly 0.0, `tie_fraction = 1.00`. The nominator has **no opinion whatsoever**. But ranking ties alphabetically put `absolute_simultaneity+no_length_contraction` – the load-bearing pair – at rank 3, and cost scored `recall@3 = 1.00`.

A signal that says nothing scored perfectly, because a sorts before p.

The fix is to break ties by a seeded shuffle, so a silent nominator scores at chance. Under it, cost on TK correctly drops to `recall@3 = 0.00` with its first load-bearing candidate at rank 7. This was corrected **before** freezing, per the preregistration's own §7, and a regression test pins it.

The general lesson transfers beyond DR1: **tie-breaking is a scoring channel, and a deterministic tie-break leaks whatever the sort key correlates with**.

4. Results

Oracle: TK has 21 candidates and 1 load-bearing; TT has 10 candidates and 4. $k = 3$.

4.1 Toy Kinematics

nominator	recall@3	regret	rank of 1st	tie frac	
weakness	1.00	0	1	0.67	
disjunctive	1.00	0	1	0.67	
size_only	0.00	1	5	0.71	control
cost	0.00	1	7	1.00	silent
random	0.00	1	19	0.05	control

Weakness puts the entangled pair at rank 1 out of 21, beating both controls. Cost is silent, as designed.

4.2 Toy Transduction

nominator	recall@3	regret	rank of 1st	tie frac	
cost	1.00	0	1	0.60	
weakness	0.67	0	1	0.60	
disjunctive	0.67	0	1	0.70	
size_only	0.67	0	1	0.60	control
random	0.67	1	2	0.10	control

Only cost beats the controls. TT **is a weak test** and this should be said plainly: 4 of 10 candidates are load-bearing, a 40% base rate, so random scores 0.67 by luck alone. TK, with 1 of 21, is the discriminating toy.

4.3 The frozen gates

gate	result
H1 – no single nominator wins both	NO_GO (TK half passes, TT half fails)
H2 – disjunction $\geq$ better single	NO_GO (TK passes, TT fails: $0.67 < 1.00$)
sanity – nominators beat controls somewhere	PASS (weakness on TK)
overall	NO_GO

5. Interpretation

H1's idea survives; H1's operationalisation does not. The binary form – "cost fires *and weakness is silent*" – fails because weakness-as-extension-size is almost never silent. Dropping nearly any non-vacuous proposition frees hypotheses, so weakness always has *some* opinion. On TT it scores 0.67, which is enough to falsify "weakness does not."

But the graded comparison is unambiguous and it points the way the hypothesis intended: **weakness 1.00 vs cost 0.00 on TK; cost 1.00 vs weakness 0.67 on TT**. Each nominator wins on a different toy, and each is *beaten* on the other. A single-objective system would take the worse of the two on one of them. The two-nominator claim should be restated in graded form – *no single nominator dominates across discovery shapes* – and DR1 supports that restatement while rejecting the binary one.

H2 fails for a specific, fixable reason. max after per-toy normalisation is a bad combiner. On TT it lets bounded_state+causal_masking – high weakness, useless – tie sequential_state_update deletions that are high cost and correct, and the tie-break then costs the disjunction a hit. A disjunction that is *worse than its own best component* is not a disjunction worth having. The

concrete lesson: combine by **sum** of normalised scores, or **lexicographically** with an explicit precedence, or gate on validity first – but not by max.

On the base rate. TT's 40% load-bearing rate makes it nearly uninformative; random gets 0.67 for free. This is the survivorship problem in miniature. A useful successor needs toys with a base rate low enough that chance performs badly – TK's 1-in-21 is the right target, and 1-in-100 would be better.

6. What DR1 licenses

Nothing beyond itself. Both preregistered hypotheses returned NO_G0, so the date-cut retrodiction on a real corpus does **not** open – that was explicitly conditioned on a G0.

What DR1 does deliver:

negatives by construction – that a corrected successor can reuse directly.

supports.

for a different program and immediately caught a real defect in this one.

1. A working harness – five slots, exhaustive oracle, execution-free nominators,
2. A specific, actionable defect in the disjunctive score, with the fix named.
3. A restatement of the two-nominator claim in the graded form the evidence
4. Independent confirmation that the E1 leakage gate generalises: it was built

Honest scope. Two authored toy systems, authored propositions, fixed vocabulary, exhaustive oracle over $|D| \leq 2$. DR1 says nothing about real corpora, nothing about vocabulary extension, and nothing about relativity or attention beyond using their shape as a template.

References

1. experiments/deletion_repair/PREREGISTRATION.md – frozen design and gates.
2. experiments/deletion_repair/results/dr1_verdict.json – the receipt behind every number above.
3. experiments/concern_gated_retrieval_e2/erratum_e1/ERRATUM.md – the leakage gate that fired here, and the program-wide defect that motivated it.
4. Lorentz (1904); Poincaré (1905); Einstein (1905); Minkowski (1908) – the dating that establishes SR as inside 1904's vocabulary and GR as outside it.
5. Bahdanau et al. (2014); Luong et al. (2015); Vaswani et al. (2017) – the dating that establishes attention as pre-existing and the *deletion* of recurrence as the contribution.

Figures

DR1 - TWO DISCOVERY SHAPES, ONE HARNESS

H1 and H2 both N0_G0 | the graded claim survives, the binary one does not

TK - kinematics

alpha cannot tell
Galilean from Lorentz

weakness 1.00
cost 0.00 (SILENT)

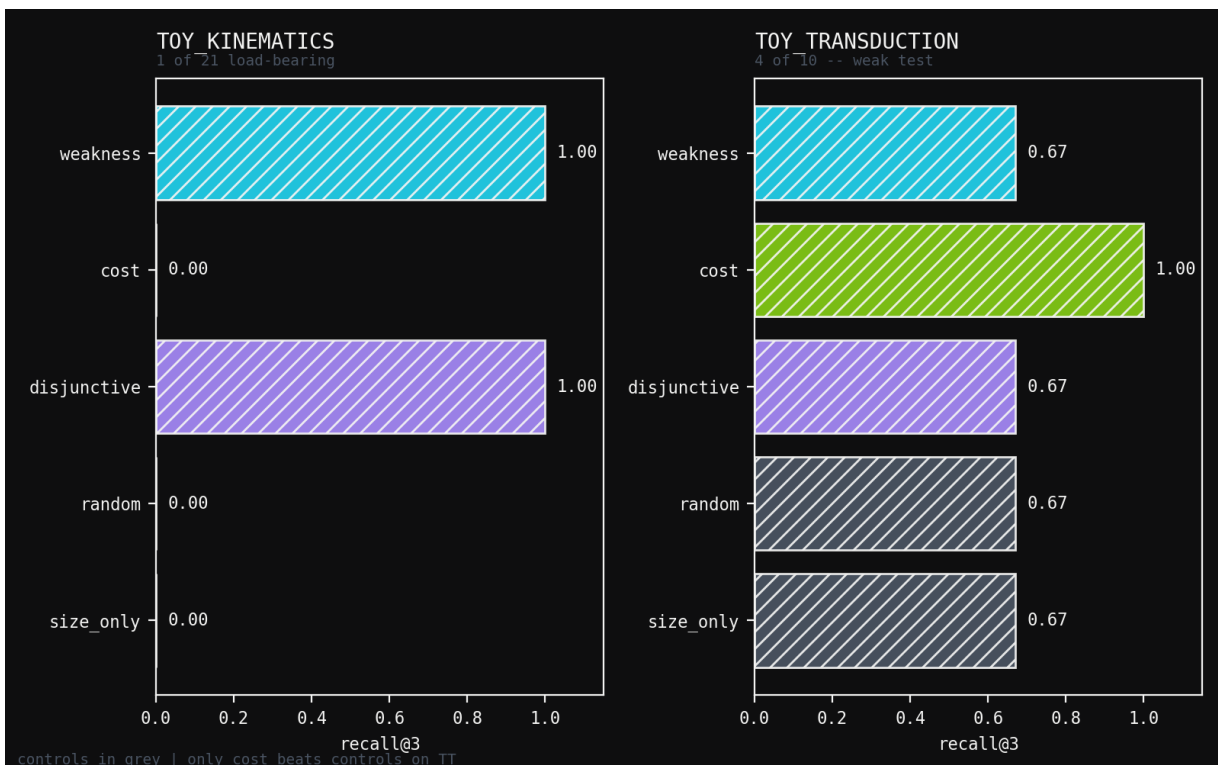
TT - transduction

depth budget, not
expressivity

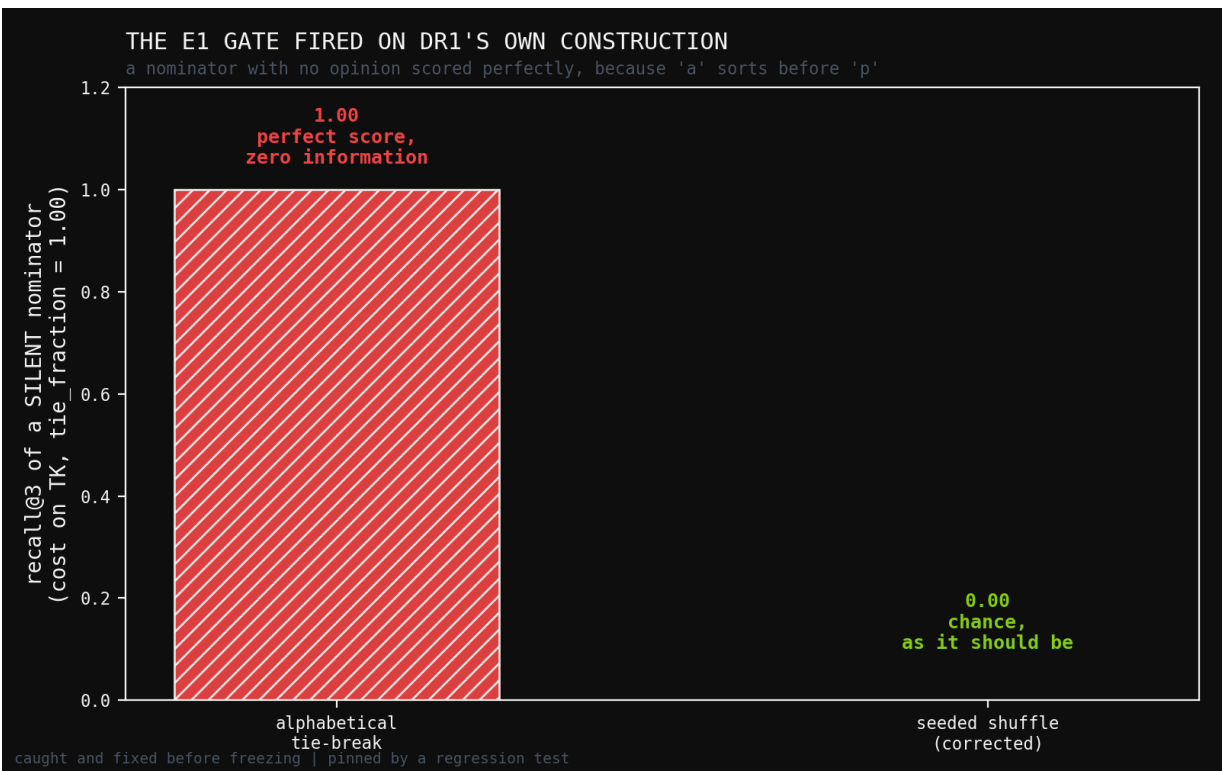
cost 1.00
weakness 0.67

each nominator wins on one toy and loses on the other

DR1 design: two toys, one harness



Nominator recall by toy



The tie-breaking leak caught during construction